

Assignment 3

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25BECCS22

3rd Sem Btech. CSE (Cyber)

Q1. Design a function that calculates the area of a rectangle. You could define default values for the width and the height but still allow the caller to provide different values when needed.

Code :

```
#include <iostream>

using namespace std ;

//q1
float rec_area(float l = 1 , float b = 1)
{
    return l*b;
}

int main()
{
    cout << "\nDefault Case \n" ;
    float l,b,area;
    cout << "Default Area : " << rec_area();

    cout << "\nEnter L : ";
    cin >> l ;
    cout << "Enter B : ";
    cin >> b ;
    area = rec_area(l,b);
    cout << "Area : " << area << endl;

    return 0;
}
```

Output :

```
Default Case
Default Area : 1
Enter L : 4.2
Enter B : 4
Area : 16.8
```

Q2. Raising a number n to a power p is the same as multiplying n by itself p times. Write a function called `power()` that takes a double value for n and an int value for p , and returns the result as a double value. Use a default argument of 2 for p , so that if this argument is omitted, the number n will be squared. Write a `main()` function that gets values from the user to test this function.

Code :

```
#include <iostream>

using namespace std ;

//q2
double power(double n, int p =2)
{
    double ans = 1;
    if (p<0)
    {
        for(int i = 0; i > p ; i--)
            ans/=n;
        return ans;
    }

    for(int i = 0; i < p ; i++)
        ans*=n;
    return ans;
}

int main()
{
    double n,ans ; int p ;

    cout << "\nFor Square\nEnter n : ";
    cin >> n ;
    ans=power(n);
    cout << "\nAns : " << ans;

    cout << endl << "For any integral power\n" << "\nEnter n : ";
    cin >> n ;
    cout << "\nEnter p : ";
    cin >> p ;
    ans=power(n,p);
    cout << "\nAns : " << ans;

    return 0;
}
```

Output :

```
For Square
Enter n : 7

Ans : 49
For any integral power

Enter n : 6

Enter p : 3

Ans : 216
```

Q3. Write a program to define function *cube()* as inline for calculating cube.

Code :

```
#include <iostream>

using namespace std ;

//q3
inline float cube(float a)
{
    return a*a*a;
}

int main()
{
    float a , ans ;
    cout << "\nEnter No. : ";
    cin >> a ;
    cout << "Cube : " << cube(a);

    return 0;
}
```

Output :

```
Enter No. : 5
Cube : 125
```

Q4. Write a program that will overload a function Volume() to compute following operations:

1. A cube (side)
2. A cuboid (length, breadth, height)
3. A sphere (radius)

Code :

```
#include <iostream>

using namespace std ;

//q4
float Volume(float a)
{
    return a*a*a;
}

double Volume(double r)
{
    return 4.0/3*22/7*r*r*r;
}

float Volume(float l, float b , float h )
{
    return l*b*h;
}

int main()
{
    int c;
    do
    {
        cout << endl << "\n\nPress\n1 : Cube\n2 : Cuboid\n3 : Sphere\nAny
other to EXIT\nEnter Choice : ";
        cin >> c ;

        if(c==1)
        {
            float a ;
            cout << "\nEnter Side : ";
            cin >> a ;
            cout << "Volume of Cube : " << Volume(a);
        }
        else if (c==2)
        {
            float l,b,h ;
            cout << "\nEnter L : ";
            cin >> l ;
            cout << "\nEnter B : ";
            cin >> b ;
            cout << "\nEnter H : ";
            cin >> h ;
            cout << "Volume of Cuboid : " << Volume(l,b,h);
        }
        else if(c==3)
        {
            double r ;
            cout << "\nEnter Radius : ";
            cin >> r ;
        }
    }
}
```

```

        cout << "Volume Of Sphere: " << Volume(r);
    }
    else
        break;
}while(c!=0);

    return 0;
}

```

Output :

```

Volume Of Sphere: 4.19048

```

Q5. Create a class containing the details of Students as details below and a main function to test the class:

Data Members(properties):

1. Name
2. Roll No
3. Degree
4. Hostel
5. CurrentCGPA

Member Function(behavior):

1. addDetails();
2. updateDetails();
3. updateCGPA();
4. updateResidenceInfo();
5. displaydetails();

Code :

```

#include <iostream>

using namespace std ;

//q5
class Students
{
    //By default private :
    string Name;
    string Degree;
    string Hostel;

    protected :

```

```

float CurrentCGPA;

public :
int RollNo;

static int studentNumber;
void addDetails()
{
    cout << endl << "\nEnter Name : ";
    cin.ignore();
    getline(cin,Name);
    cout << endl << "Enter Roll No. : ";
    cin >> RollNo;
    cout << endl << "Enter Degree : ";
    cin.ignore();
    getline(cin,Degree);
    cout << endl << "Enter Hostel : ";
    cin.ignore();
    getline(cin,Hostel);
    cout << endl << "Enter Current CGPA : ";
    cin >> CurrentCGPA;
    studentNumber++;
}
void updateDetails()
{
    int x;
    cout << endl << "\nUpdate MENU\n\n1 : Name\n2 : Roll No.\n3 :
Degree\n4 : Hostel\n5 : Current CGPA\nChoose : ";
    cin >> x;
    if (x==1)
    {
        cout << endl << "\nEnter New Name : ";
        cin.ignore();
        getline(cin,Name);
    }
    else if (x==2)
    {
        cout << endl << "Enter new Roll No. : ";
        cin >> RollNo;
    }
    else if (x==3)
    {
        cout << endl << "Enter new Degree : ";
        cin.ignore();
        getline(cin,Degree);
    }
    else if (x==4)
    {
        cout << endl << "Enter new Hostel : ";
        cin.ignore();
        getline(cin,Hostel);
    }
    else if (x==5)
    {
        cout << endl << "Enter new Current CGPA : ";

```

```

        cin >> CurrentCGPA;
    }
    else
        cout << "\nPlease Enter No. from 1 to 5 only !\n";
}

inline void updateCGPA();

void updateResidentInfo()
{
    cout << endl << "Enter new Hostel : ";
    cin.ignore();
    getline(cin, Hostel);
}

void displaydetails()
{
    cout << "\n\n~ ~ ~ ~ ~ Details ~ ~ ~ ~ ~\n";
    cout << "Name : " << Name << "\nRoll No. : " << RollNo << "\nDegree : " << Degree << "\nHostel : " << Hostel << "\nCurrent CGPA : " << CurrentCGPA;
}

};

void Students::updateCGPA()
{
    cout << endl << "Enter new Current CGPA : ";
    cin >> CurrentCGPA;
}

int Students::studentNumber = 0;

int main()
{
    int choic, stu;
    Students a , b ;

    cout << endl << "\nEnter Details of Student A : ";
    a.addDetails();
    cout << endl << "\nEnter Details of Student B : ";
    b.addDetails();

    do
    {
        cout << "\nPress\n1 : Update Details\n2 : Update CGPA\n3 : Update Resident Info\n4 : Display Details\n0 : EXIT\nChoose : " ;
        cin >> choic ;
        if(choic==0)
            break;
        cout << "\nPress\n1 : Student A\n2 : Student B\nChoose :";
        cin >> stu;
        if (choic==1 && stu==1)
            a.updateDetails();
        else if (choic==2 && stu==1)

```

```

        a.updateCGPA();
        else if (choic==3 && stu==1)
            a.updateResidentInfo();
        else if (choic==4 && stu==1)
            a.displaydetails();

        else if (choic==1 && stu==2)
            b.updateDetails();
        else if (choic==2 && stu==2)
            b.updateCGPA();
        else if (choic==3 && stu==2)
            b.updateResidentInfo();
        else if (choic==4 && stu==2)
            b.displaydetails();
    }while (choic!=0);

    cout << "\nExtra Experimentatin\n";
    // cout << a.CurrentCGPA; // Would give error as it's protected , cannot
    be accessed directly in main
    cout << endl << "Displaying Roll No. of B directly : " << b.RollNo;
    // cout << endl << a.Name; // Would give error as it's protected , cannot
    be accessed directly in main
    cout << "\nCount of students through static : " << a.studentNumber;

    return 0;
}

```

Output :

Enter Details of Student A :	Press
	1 : Student A
Enter Name : Darsh Dhawan	2 : Student B
	Choose :2
Enter Roll No. : 67	
Enter Degree : Btech CSE Cyber	~~~~~ Details ~~~~~
	Name : Aarav Raina
Enter Hostel : BRS	Roll No. : 3
	Degree : Btech AI/ML
Enter Current CGPA : 8.88	Hostel : PM
	Current CGPA : 6.67
	Press
	1 : Update Details
Enter Details of Student B :	2 : Update CGPA
	3 : Update Resident Info
Enter Name : Aarav Raina	4 : Display Details
	0 : EXIT
Enter Roll No. : 3	Choose : 2
Enter Degree : Btech AI/ML	Press
	1 : Student A
Enter Hostel : SPM	2 : Student B
	Choose :2
Enter Current CGPA : 6.67	
	Enter new Current CGPA : 9.99
Press	
1 : Update Details	Press
2 : Update CGPA	1 : Update Details
3 : Update Resident Info	2 : Update CGPA
4 : Display Details	3 : Update Resident Info
0 : EXIT	4 : Display Details
Choose : 4	0 : EXIT
	Choose : 4

```
Press
1 : Student A
2 : Student B
Choose :2

~~~~~ Details ~~~~~
Name : Aarav Raina
Roll No. : 3
Degree : Btech AI/ML
Hostel : PM
Current CGPA : 9.99
Press
1 : Update Details
2 : Update CGPA
3 : Update Resident Info
4 : Display Details
0 : EXIT
Choose : 1

Press
1 : Student A
2 : Student B
Choose :2

Update MENU

1 : Name
2 : Roll No.
3 : Degree
4 : Hostel
5 : Current CGPA
Choose : 4

Enter new Hostel : BRS

Press
1 : Update Details
2 : Update CGPA
3 : Update Resident Info
4 : Display Details
```

```
0 : EXIT
Choose : 4

Press
1 : Student A
2 : Student B
Choose :2

~~~~~ Details ~~~~~
Name : Aarav Raina
Roll No. : 3
Degree : Btech AI/ML
Hostel : BRS
Current CGPA : 9.99
Press
1 : Update Details
2 : Update CGPA
3 : Update Resident Info
4 : Display Details
0 : EXIT
Choose : 0

Extra Experimentatin

Displaying Roll No. of B directly : 3
Count of students through static : 2
```

Q6. Design and implement a **Student** class in C++ to develop a simple student performance management system.

The class should contain the following data members:

- name – stores the student's name.
- roll_number – stores the student's unique roll number.
- marks – stores the marks obtained by the student in multiple subjects.

Implement the following member functions:

1. calculate_average() – calculates and returns the student's average marks.
2. get_grade() – determines and returns the student's letter grade based on the average marks using an appropriate grading scheme.

Create objects for **multiple students (5)**, ensuring that each student has a unique name, roll number, and set of marks. Invoke the appropriate member functions for every student and display an individual **performance report** containing the student's name, roll number, average marks, and grade.

Code :

```
#include <iostream>
#include <string>

using namespace std;

class Student
{
    // by default private :
    string name;
    float marks[5];
    string subjects[5]={"Python","Data Science","C
Programming","DSA","Cryptography"};
public :
    int roll_number;

    void add_details()
    {
        cout << endl << "Name : ";
        getline(cin,name);

        cout << endl << "Roll No. : ";
        cin >> roll_number;

        for(int i=0 ; i<5 ; i++)
        {
            cout << endl << subjects[i] << " : ";
            cin >> marks[i];
        }
        cin.ignore();
    }

    float calculate_average()
    {
        float avg=0;

        for(float i : marks)
```

```

        avg+=i;

        avg/=5;
        return avg;
    }

    char get_grade()
    {

        float avg = calculate_average();
        if(avg >= 90 )
            return 'O';

        else if(avg >= 80 )
            return 'A';

        else if(avg >= 70 )
            return 'B';

        else if(avg >= 60 )
            return 'C';

        else if(avg >= 45 )
            return 'D';

        else
            return 'F';
    }

    void performance_report()
    {
        cout << endl << " ~ ~ ~ ~ ~ Performance Report ~ ~ ~ ~ ~ ";
        cout << " \n\nName : " << name;
        cout << " \nRoll No. : " << roll_number;
        cout << " \nAverage Marks : " << calculate_average();
        cout << " \nGrade : " << get_grade();
    }
};

Student students[5];

int main()
{
    cout <<endl << " ~ ~ ~ ~ ~ Student Performance Management System ~ ~ ~ ~ ~ \n\n";

    cout << "Enter Details of Students \n\n";

    for (Student &i : students )
    {
        cout << endl ;
        i.add_details();
    }

    for (int i=0 ; i<5 ; i++ )

```

```

{
    for(int j=0 ; j<5 ; j++)
    {
        if (i!=j && students[i].roll_number == students[j].roll_number )
        {
            cout << "\n\nRoll Numbers can't be Duplicated";
            return 1;
        }
    }
}

for (Student &i : students )
{
    i.performance_report();
}

return 0;
}

```

Output :

```

~ ~ ~ ~ ~ Student Performance Management System ~ ~ ~ ~ ~
Enter Details of Students

Name : Darsh Dhawan
Roll No. : 22
Python : 78
Data Science : 89
C Programming : 90
DSA : 90
Cryptography : 100

Name : Aarav
Roll No. : 3
Python : 78
Data Science : 87
C Programming : 67
DSA : 89
Cryptography : 67

~ ~ ~ ~ ~ Performance Report ~ ~ ~ ~ ~
Name : Darsh Dhawan
Roll No. : 22
Average Marks : 89.4
Grade : A
~ ~ ~ ~ ~ Performance Report ~ ~ ~ ~ ~
Name : Aarav
Roll No. : 3
Average Marks : 77.6
Grade : B
~ ~ ~ ~ ~ Performance Report ~ ~ ~ ~ ~
Name : Laalu Yadav
Roll No. : 69
Average Marks : 35.8
Grade : F
~ ~ ~ ~ ~ Performance Report ~ ~ ~ ~ ~
Name : Mohit Tyagi
Roll No. : 23
Average Marks : 11.4
Grade : F
~ ~ ~ ~ ~ Performance Report ~ ~ ~ ~ ~
Name : Mohak Mangal
Roll No. : 65
Average Marks : 76.2
Grade : B

```

Q7. Create a code snippet that illustrates the following:

1. calling of private member functions inside public member function
2. Access private data members inside public member functions.

Code :

```
#include <iostream>

using namespace std;

class theclass
{
    //by default private :

    static int var2;
    void private_fn()
    {
        cout << "\nHi ! I'm in Private member function";
    }
public:
    static void public_fn()
    {
        cout << "\nYo ! I'm inside Public Member Function.";
        cout << "\nHere I accessed private data member inside Public
Member fn \n";
        cout << "var2 is : " << var2;
    }
};

int theclass::var2=2;

int main()
{
    theclass::public_fn();
    return 0;
}
```

Output :

```
Yo ! I'm inside Public Member Function.
Here I accessed private data member inside Public Member fn
var2 is : 2
```

~~~~~ E N D ~~~~~